A-10 Data Visualization III Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., https://archive.ics.uci.edu/ml/datasets/Iris). Scan the dataset and give the inference as:

1. List down the features and their types (e.g., numeric, nominal) available in the dataset.
2. Create a histogram for each feature in the dataset to illustrate the feature distributions.
3. Create a box plot for each feature in the dataset.
4. Compare distributions and identify outliers

In [1]:
```
import seaborn as sns
import pandas as pd
import matplotlib.pyplot as plt
```

In [2]:
```
df=sns.load_dataset('iris')
df.head()
```

Out[2]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

In [3]:
```
df.tail()
```

Out[3]:

	sepal_length	sepal_width	petal_length	petal_width	species
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

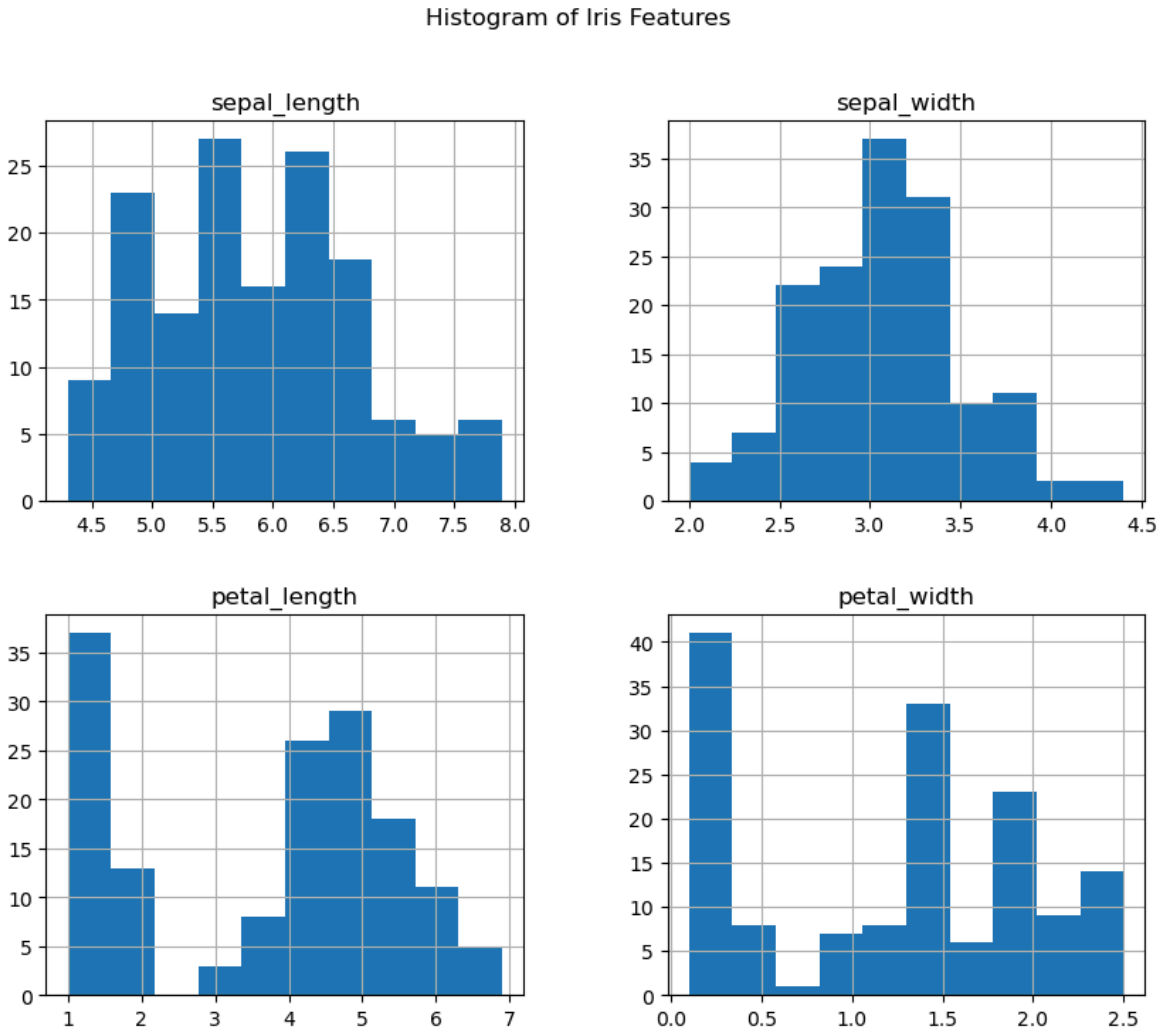
In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [5]:
```
df.isnull().sum()
```

Out[5]:
```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

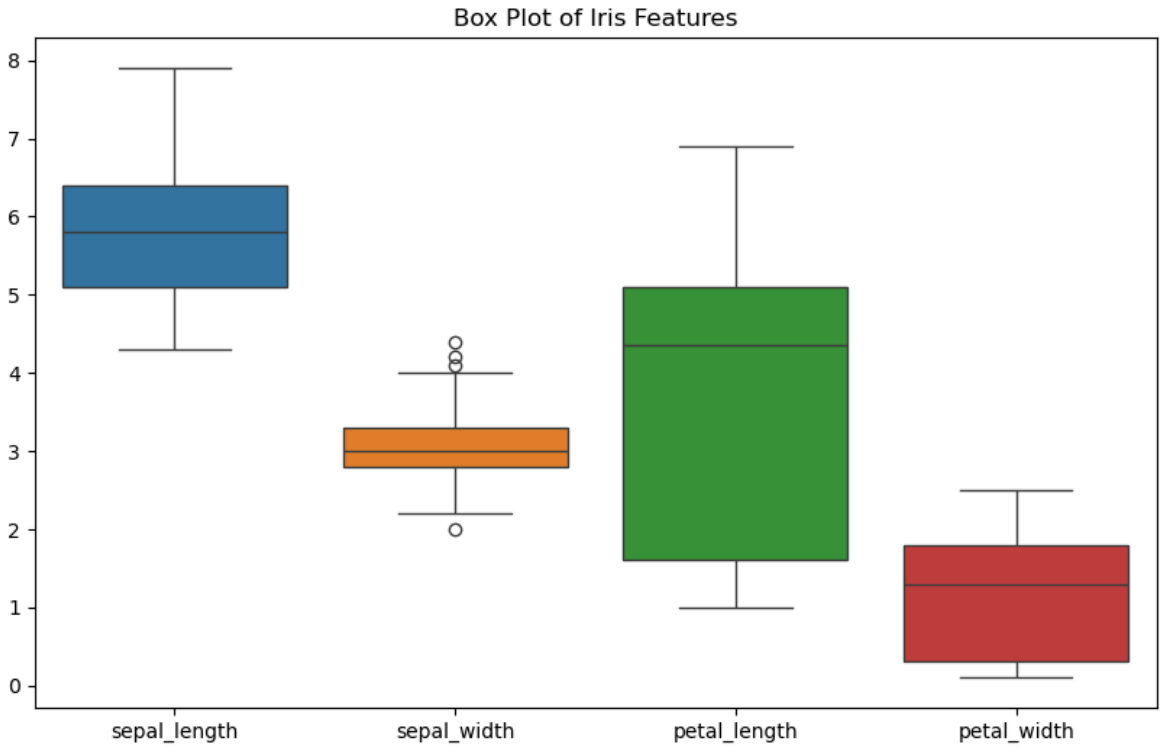
In [6]:
```
df.hist(figsize=(10,8))
plt.suptitle("Histogram of Iris Features")
plt.show()
```



Inference : Petal length and petal width distributions are distinctly different across species, making them strong classification features.Most features follow approximately

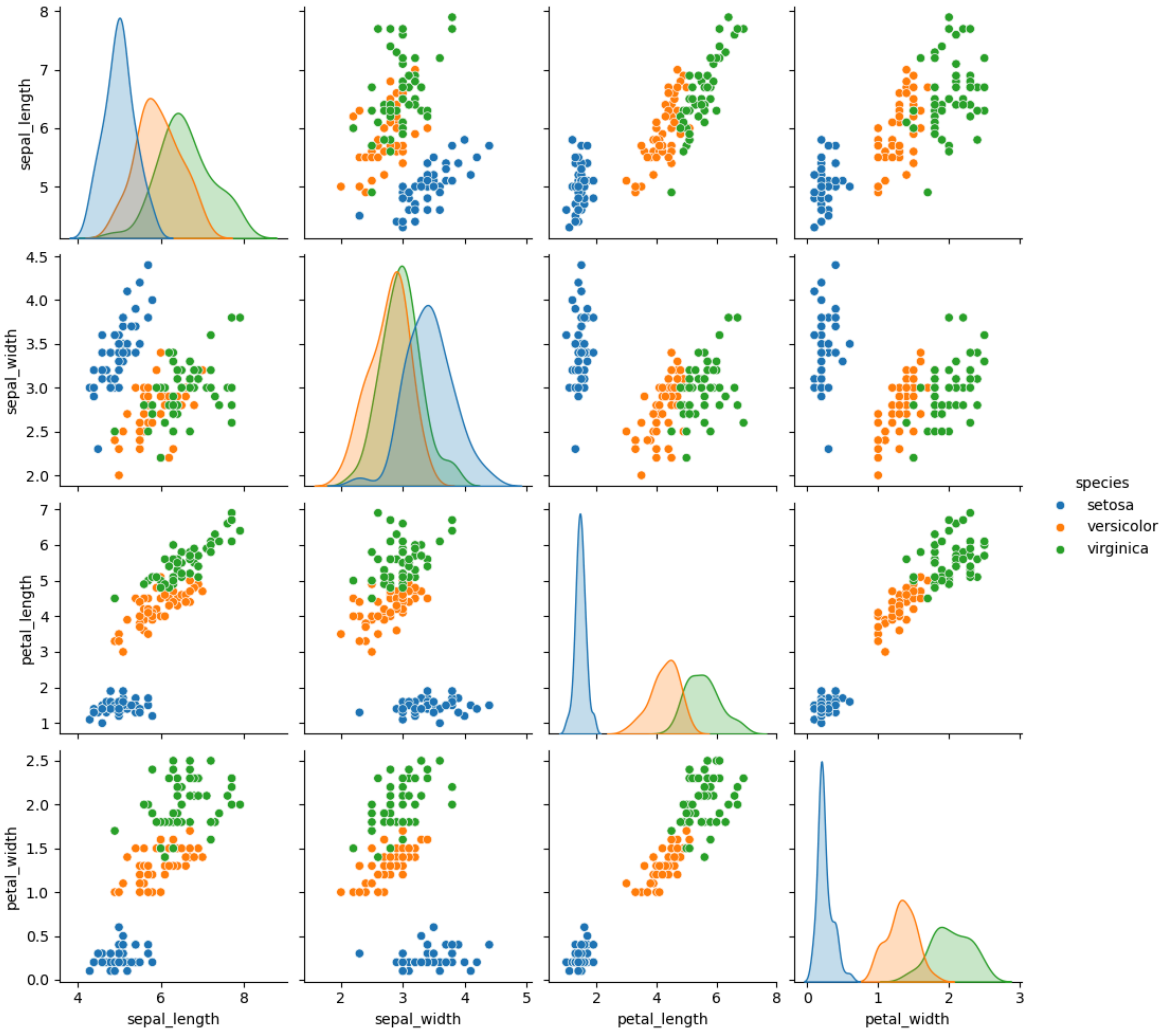
normal distribution. Slight skewness observed in sepal width

In [7]:
```
plt.figure(figsize=(10,6))
sns.boxplot(data=df)
plt.title("Box Plot of Iris Features")
plt.show()
```



Inference : Some outliers are observed in sepal width and petal width, visible as points beyond whiskers in box plots.

In [8]:
```
sns.pairplot(df, hue='species')
plt.show()
```



Features with types sepal_length : numeric (continuous) sepal_width : numeric (continuous) petal_length: numeric (continuous) petal_width : numeric (continuous) species: categorical (setosa, versicolor, virginica)